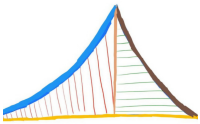
CBSE 2026
12th Class Maths
Solutions (65-5-1)



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9–12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share. License:
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First manual appeared in January 2018
First edition published on July 10, 2024
In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CBSE 2026 12th class maths solutions (65-5-1)

FORMULAE

1.1.1. Matrix identities. If $A^2 = I$, then $A^{-1} = A$. If $A^2 = A$, then A is idempotent and $A^n = A$ for every integer $n \geq 1$. Also $A^{-1} = \operatorname{adj}(A)/\det A$ when $\det A \neq 0$.

1.1.2. Continuity, derivatives, and extrema. A function is continuous at a when $\lim_{x\to a} f(x) = f(a)$. A differentiable function can have an interior local extremum only where $f'(x) = 0$. For $x > 0$,

$$\frac{d}{dx}x^x = x^x(1 + \ln x). \tag{1.1.2.1}$$

1.1.3. Trigonometric identity and integral.

$$1 + \cos 2x = 2\cos^2 x, \qquad \int \sec x\,dx = \ln|\sec x + \tan x| + C. \tag{1.1.3.1}$$

Thus, on intervals where $\cos x > 0$, $\int \frac{dx}{\sqrt{1+\cos 2x}} = \frac{1}{\sqrt{2}}\ln|\sec x + \tan x| + C$.

1.1.4. Area of an ellipse. For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the semi-axes are a, b and the enclosed area is πab.

1.1.5. Linear programming. For a linear objective $Z = ax + by$, find its maximum or minimum by evaluating Z at each vertex of the feasible region.

1.1.6. Conditional probability and Bayes' theorem.

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{1.1.6.1}$$

If $\{A_i\}$ partitions the sample space, $P(A_i \mid B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$.

1.1.7. Vectors.

$$\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta, \qquad \text{Area}(\triangle ABC) = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|. \tag{1.1.7.1}$$

For unit vectors, $|\mathbf{a} - \mathbf{b}|^2 = 2 - 2\mathbf{a}\cdot\mathbf{b}$.

1.1.8. Angle between lines. If their direction vectors are $\mathbf{d}_1, \mathbf{d}_2$, then $\cos\theta = \frac{|\mathbf{d}_1\cdot\mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}$ for the acute angle.

1.1.9. Distance between skew lines and from a point to a line.

$$d = \frac{|(\mathbf{a}_2 - \mathbf{a}_1)\cdot(\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}. \tag{1.1.9.1}$$

For a point P and a line $\mathbf{r} = \mathbf{a} + t\mathbf{d}$, the distance is

$$d(P, \ell) = \frac{|(\mathbf{p} - \mathbf{a}) \times \mathbf{d}|}{|\mathbf{d}|}. \tag{1.1.9.2}$$

1.1.10. First-order linear differential equations. For $x' + P(y)x = Q(y)$, the integrating factor is $\mu(y) = e^{\int P(y)\,dy}$ and $(\mu x)' = \mu Q$.

1.1.11. Relations and functions. If finite sets A, B have m, n elements, respectively, the number of relations from A to B is 2^{mn} and the number of functions from A to B is n^m.

1.1.12. Linear systems in matrix form. A system of linear equations can be written $AX = B$; if $\det A \neq 0$, then $X = A^{-1}B$.

1.1.13. Area of a triangle from coordinates. For vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3), $[ABC] = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.

SOLVED QUESTIONS

5.1.1 If matrix $A = \begin{bmatrix} -p & q \\ r & p \end{bmatrix}$ is such that $A^2 = I$, then:
a) $1+p^2+qr=0$ b) $1-p^2-qr=0$ c) $1-p^2+qr=0$ d) $1+p^2-qr=0$

Solution:
Squaring the matrix gives

$$A^2 = \begin{bmatrix} p^2+qr & 0 \\ 0 & p^2+qr \end{bmatrix} = (p^2+qr)I. \tag{1.1.1}$$

Since $A^2 = I$, we have $p^2 + qr = 1$, or $1 - p^2 - qr = 0$. The correct answer is **b)**.

5.1.2 If A is a square matrix such that $A^2 = A$, then $(A-I)^3 - A$ is equal to:
a) I b) $-I$ c) A d) A^2

Solution:
Using $A^2 = A$,

$$(A-I)^2 = A^2 - 2A + I = I - A, \tag{1.1.1}$$

and hence

$$(A-I)^3 = (I-A)(A-I) = A - I.$$

Therefore $(A-I)^3 - A = (A-I) - A = -I$. The correct answer is **b)**.

5.1.4 Let

$$A = \begin{bmatrix} 0 & -3 & 4 \\ 1 & 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} -3 & 0 & 1 \\ 2 & 4 & 0 \end{bmatrix}.$$

If $A + B + C = O$, then matrix C is:
a) $\begin{bmatrix} -3 & -3 & 5 \\ 3 & 4 & 2 \end{bmatrix}$ b) $\begin{bmatrix} 3 & 3 & 5 \\ -3 & -4 & -2 \end{bmatrix}$ c) $\begin{bmatrix} 3 & 3 & -5 \\ -3 & -4 & -2 \end{bmatrix}$ d) $\begin{bmatrix} -3 & -3 & -5 \\ 3 & 4 & 2 \end{bmatrix}$

Solution:
$C = -(A+B)$. Adding the matrices gives $A + B = \begin{bmatrix} -3 & -3 & 5 \\ 3 & 4 & 2 \end{bmatrix}$, so

$$C = \begin{bmatrix} 3 & 3 & -5 \\ -3 & -4 & -2 \end{bmatrix}. \tag{1.1.1}$$

The answer is **c)**.

5.1.5 If A is a non-singular matrix, which statement is not true?
a) $\operatorname{adj} A$ is singular b) $(\operatorname{adj} A)^{-1} = \operatorname{adj}(A^{-1})$ c) $|A| \neq 0$ d) A^{-1} exists

Solution:
For a non-singular matrix, $|A| \neq 0$ and A^{-1} exists. Also $\operatorname{adj} A = |A|A^{-1}$ is non-singular. Therefore the false statement is that $\operatorname{adj} A$ is singular, option **a)**.

The accompanying plot uses a sample matrix family to illustrate that singularity occurs only when its determinant is zero.

(1.1.1)

codes/5.1.5.py

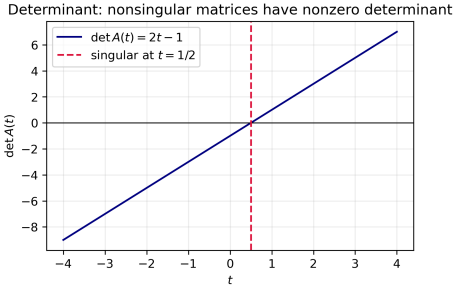


Fig. 1: Determinant of a representative matrix family.

5.1.6 If

$$f(x) = \begin{cases} \dfrac{x^2 - 4x - 5}{x + 1}, & x \neq -1, \\ k, & x = -1, \end{cases}$$

is continuous at $x = -1$, find k.

Solution:
For $x \neq -1$, $f(x) = \frac{(x-5)(x+1)}{x+1} = x - 5$. Thus $\lim_{x \to -1} f(x) = -6$, and continuity requires $k = -6$, option **d)**. The code plots the resulting continuous function.

(1.1.2)

codes/5.1.6.py

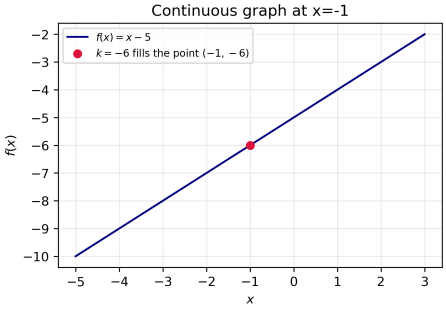


Fig. 2: The continuous extension at $x = -1$.

5.1.7 The triangle with vertices $A(3,1)$, $B(-2,1)$ and $C(0,k)$ has area 5 square units. Find k.

a) $3,1$ b) $-1,3$ c) $-1,2$ d) $0,2$

Solution:

The base AB has length 5 and the perpendicular height is $|k-1|$. Hence $\frac{1}{2}(5)|k-1| = 5$, giving $|k-1| = 2$, so $k = -1$ or $k = 3$, option **b)**. Both triangles are plotted by the code.

(1.1.13)

codes/5.1.7.py

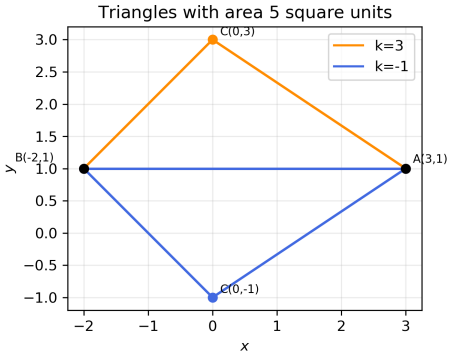


Fig. 3: Both possible triangles with area 5.

5.1.9 Find the absolute minimum value of $f(x) = (x-2)^2 + 5$ on $[-3,2]$.

a) -3 b) 2 c) 5 d) 30

Solution:

Since $(x-2)^2 \geq 0$ with equality at the included endpoint $x = 2$, the absolute minimum is $f(2) = 5$, option **c)**. The code performs a bounded numerical minimization and plots the function.

(1.1.2)

codes/5.1.9.py

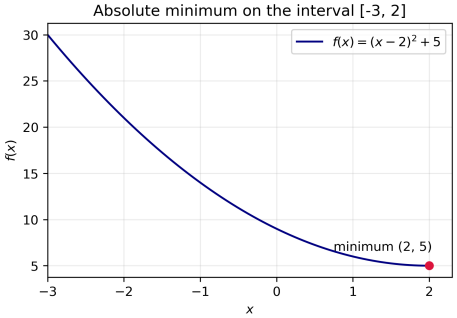


Fig. 4: The minimum occurs at $x = 2$.

5.1.10 Evaluate $\int \frac{1}{\sqrt{1+\cos 2x}}\,dx$.

a) $\log\cos x + C$ $\tan x| + C$ $\tan x| + C$

b) $\frac{1}{\sqrt{2}}\log|\sec x +$ c) $\frac{1}{\sqrt{2}}\log|\sec x -$ d) $\log\sin 2x + C$

Solution:

Using $1+\cos 2x = 2\cos^2 x$, the integrand is $\frac{1}{\sqrt{2}}\sec x$ on intervals where $\cos x > 0$. Therefore

$$\int \frac{dx}{\sqrt{1+\cos 2x}} = \frac{1}{\sqrt{2}}\log|\sec x + \tan x| + C, \tag{1.1.3}$$

which is option **b)**.

5.1.12 An ant crawls along the line $y = 2x - 4$. Find the area bounded by this line, the y-axis, the x-axis, and $x = 1$.

a) 1 sq. unit b) 3 sq. units c) 2 sq. units d) 4 sq. units

Solution:

For $0 \le x \le 1$, the line is below the x-axis, so the area is $\int_0^1 (4-2x)\,dx = 3$ square units, option **b)**. The code plots the line and bounded region.

codes/5.1.12.py

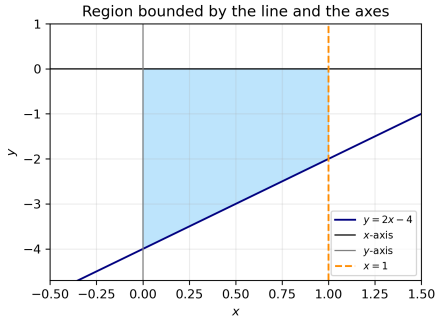


Fig. 5: Area enclosed by $y = 2x - 4$, the axes, and $x = 1$.

5.1.15 The feasible-region corner points are $(0, 0), (0, 40), (20, 40), (60, 20), (60, 0)$. For $Z = 4x + 3y$, find the maximum value.

a) 200 b) 300 c) 240 d) 120

Solution:

At the listed corners, Z takes values $0, 120, 200, 300, 240$, respectively. Thus the maximum is 300 at $(60, 20)$, option **b)**. The code plots the feasible polygon and labels its corner values.

(1.1.5)

codes/5.1.15.py

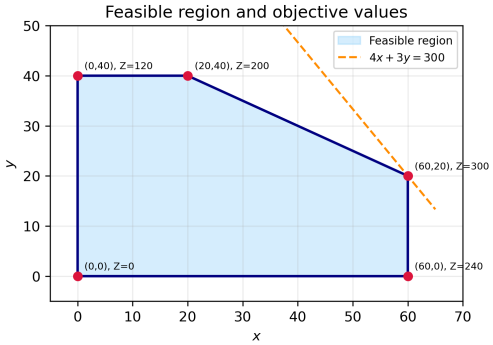


Fig. 6: Feasible region and objective values at its corners.

5.1.18 Given $3P(A) = P(B) = \frac{3}{5}$ and $P(A \mid B) = \frac{2}{3}$, find $P(A \cup B)$.

a) $\frac{3}{5}$ b) $\frac{1}{5}$ c) $\frac{2}{15}$ d) $\frac{2}{5}$

Solution:

As printed, $P(A) = \frac{1}{5}$ and $P(B) = \frac{3}{5}$. The conditional probability gives $P(A \cap B) = P(A \mid B)P(B) = \frac{2}{5}$, so direct substitution into the union formula yields

$P(A \cup B) = \frac{1}{5} + \frac{3}{5} - \frac{2}{5} = \frac{2}{5}$ (option **d**)). However, these data are inconsistent: an intersection cannot have probability $\frac{2}{5}$ when $P(A) = \frac{1}{5}$. Thus option d) is the formal calculation from the printed values, but the question appears to contain a typo.

(1.1.6)

5.1.20 Consider minimizing $Z = x + 2y$ subject to $2x + y \geq 3$, $x + 2y \geq 6$, and $x, y \geq 0$. The assertion says the minimum occurs at infinitely many points and the feasible-region corners are $(0, 3)$ and $(6, 0)$. The reason says that if two corners give the same minimum, every point on the segment between them also gives that minimum. Select the correct assertion-reason choice.

Solution:
The two corners both give $Z = 6$. Since Z is linear, every point on the segment joining them also gives $Z = 6$, and the constraints are satisfied along that segment. Both statements are true, and the reason explains the assertion: option **a**). The code plots the constraint boundaries and the optimal segment.

(1.1.5)

codes/5.1.20.py

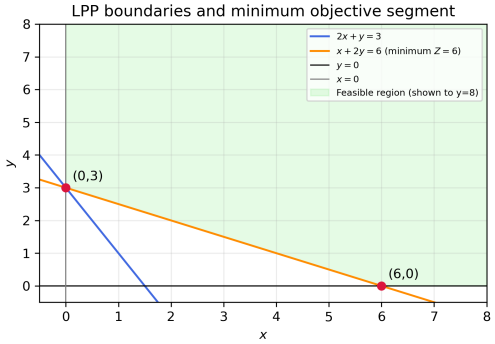


Fig. 7: Constraints and the segment of optimal points.

5.1.22.1 Differentiate x^x with respect to $x \log x$.

Solution:
Let $t = x \log x$ and $y = x^x$. Then $\log y = x \log x = t$, so $y = e^t$. Hence $\frac{dy}{d(x \log x)} = \frac{dy}{dt} = e^t = x^x$.

(1.1.2)

5.1.22.2 If $y = P \cos ux + Q \sin ux$, show that $\frac{d^2y}{dx^2} + u^2 y = 0$.

Solution:

Differentiating twice gives $y'' = -Pu^2\cos ux - Qu^2\sin ux = -u^2y$. Therefore $y'' + u^2y = 0$.

(1.1.2)

5.1.23 Find the values of x for which $f(x) = \frac{x-3}{x+1}$, $x \neq -1$, is increasing.

Solution:
$f'(x) = \frac{(x+1)-(x-3)}{(x+1)^2} = \frac{4}{(x+1)^2} > 0$ for every $x \neq -1$. Thus f is increasing on $(-\infty, -1)$ and $(-1, \infty)$. The code plots the function and its asymptote.

(1.1.2)

codes/5.1.23.py

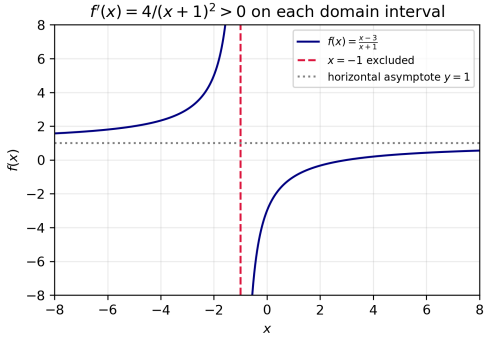


Fig. 8: The rational function increases on each interval of its domain.

5.1.24.1 Three bees fly along vectors $\mathbf{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\mathbf{b} = 4\hat{j} - 2\hat{k}$, and $\mathbf{c} = 3\hat{i} + 2\hat{k}$. Find λ such that $\mathbf{a} + \lambda\mathbf{b}$ is perpendicular to $\mathbf{c}$.

Solution:
Perpendicularity requires $(\mathbf{a} + \lambda\mathbf{b}) \cdot \mathbf{c} = 0$. Since $\mathbf{a} \cdot \mathbf{c} = 8$ and $\mathbf{b} \cdot \mathbf{c} = -4$, we get $8 - 4\lambda = 0$, hence $\boxed{\lambda = 2}$.

(1.1.7)

5.1.24.2 For non-collinear points A, B, C, $\overrightarrow{AB} = \hat{i} + 2\hat{j} - \hat{k}$ and $\overrightarrow{AC} = 2\hat{i} - 3\hat{j}$. Find the area of $\triangle ABC$.

Solution:
The area is $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$. Their cross product is $(-3, -2, -7)$, whose magnitude is $\sqrt{62}$. Therefore the area is $\boxed{\frac{\sqrt{62}}{2}}$ square units. The code plots the triangle in 3D and evaluates the cross product.

(1.1.7)

codes/5.1.24.2.py

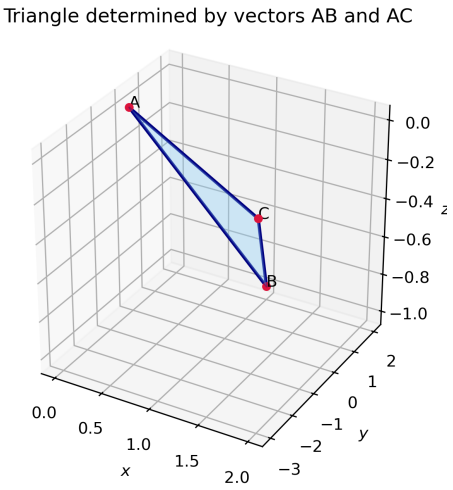


Fig. 9: Triangle formed by the two position-difference vectors.

5.1.25 Find the angle between the lines $\frac{x-2}{3} = \frac{y+5}{2} = \frac{1-z}{-6}$ and $\frac{x-7}{1} = \frac{y}{2} = \frac{6-z}{-2}$.

Solution:
Direction vectors are $(3, 2, 6)$ and $(1, 2, 2)$. Their dot product is 19 and their magnitudes are 7 and 3. Thus the acute angle is $\theta = \cos^{-1}(19/21) \approx 25.2°$. The code plots both 3D lines and calculates the angle using their direction vectors. *(1.1.8)*

codes/5.1.25.py

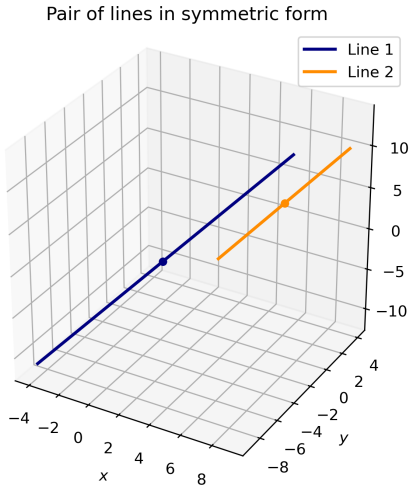


Fig. 10: The two lines in three-dimensional space.

5.1.28 Solve the differential equation $(x + 2y^3)\,dy = y\,dx$.

Solution:
Treat x as a function of y: $\frac{dx}{dy} - \frac{x}{y} = 2y^2$. The integrating factor is $1/y$, so $\frac{d}{dy}(x/y) = 2y$. Integration gives $x/y = y^2 + C$, or $\boxed{x = y^3 + Cy}$.

(1.1.10)

5.1.30.2 If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors, prove $|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2 \leq 9$.

Solution:
Using $|\mathbf{a}| = |\mathbf{b}| = |\mathbf{c}| = 1$,

$$|\mathbf{a} - \mathbf{b}|^2 + |\mathbf{b} - \mathbf{c}|^2 + |\mathbf{c} - \mathbf{a}|^2 = 9 - |\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 \leq 9. \qquad (1.1.7)$$

Equality is possible when the vectors sum to zero, for example when three coplanar unit vectors are 120° apart.

5.1.31.1 A die is rolled with $A = \{1, 2, 5\}$, $B = \{3, 5\}$ and $C = \{2, 3, 4, 5\}$. Find (i) $P(A \mid C)$ and $P(C \mid A)$; (ii) $P(A \cap B \mid C)$ and $P(A \cup B \mid C)$.

Solution:
$A \cap C = \{2, 5\}$ and $C \cap A = \{2, 5\}$, so $P(A \mid C) = 2/4 = 1/2$ and $P(C \mid A) = 2/3$. Also $A \cap B \cap C = \{5\}$, giving $P(A \cap B \mid C) = 1/4$. Finally $(A \cup B) \cap C = \{2, 3, 5\}$, so $P(A \cup B \mid C) = 3/4$.

(1.1.6)

5.1.31.2 A box contains cards numbered 1 to 6. Two cards are drawn with replacement. Let A be the event that their sum is 10 and B the event that the first number is

not 4. Find $P(A \cap B)$ and determine whether A and B are independent.

Solution:
There are 36 ordered outcomes. $A = \{(4,6),(5,5),(6,4)\}$ and $A \cap B = \{(5,5),(6,4)\}$, so $P(A \cap B) = 2/36 = 1/18$. But $P(A)P(B) = (3/36)(30/36) = 5/72 \neq 1/18$. Thus the events are not independent.

(1.1.6)

5.1.32 Four apples, three oranges and two bananas cost |60; two apples, four oranges and six bananas cost |90; six apples, two oranges and three bananas cost |70. Find the price of one fruit of each kind using the matrix method.

Solution:
Let the prices be a, o, b. The system is $4a + 3o + 2b = 60$, $2a + 4o + 6b = 90$, $6a + 2o + 3b = 70$. Solving the corresponding matrix equation gives $a = 5$, $o = 8$, $b = 8$. Thus an apple costs |5, an orange |8, and a banana |8.

(1.1.1, 1.1.12)

5.1.33.1 If $y\sqrt{x^2+1} = \log(\sqrt{x^2+1} - x)$, show that $(x^2+1)\frac{dy}{dx} + xy + 1 = 0$.

Solution:
Differentiate implicitly. The derivative of the right side is $-1/\sqrt{x^2+1}$. The left derivative is $y'\sqrt{x^2+1} + xy/\sqrt{x^2+1}$. Multiplying by $\sqrt{x^2+1}$ gives $(x^2+1)y' + xy = -1$, which is the required result.

(1.1.2)

5.1.33.2 Find the derivative with respect to x of $x^{\cot x} + \frac{2x^2-3}{2x^2-x+2}$.

Solution:
Logarithmic differentiation gives $\frac{d}{dx}x^{\cot x} = x^{\cot x}(\frac{\cot x}{x} - \log x \csc^2 x)$. By the quotient rule, the derivative of the rational term is $\frac{-2x^2+20x-3}{(2x^2-x+2)^2}$. Add these two expressions for the required derivative.

(1.1.2)

5.1.34 Sketch $\{(x,y) : 100x^2 + 25y^2 = 2500\}$ and find its enclosed area using integration.

Solution:
The equation is $x^2/25 + y^2/100 = 1$, an ellipse with semi-axes 5 and 10. By symmetry, its area is

$$4\int_0^5 10\sqrt{1 - x^2/25}\,dx = 50\pi. \qquad (1.1.4)$$

The code plots the ellipse and computes the enclosed area.

codes/5.1.34.py

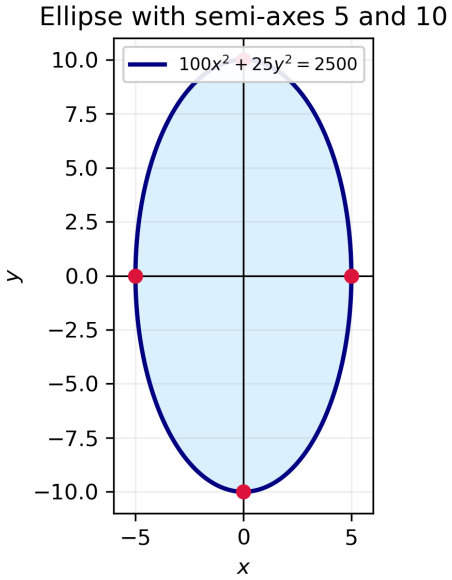


Fig. 11: Ellipse $x^2/25 + y^2/100 = 1$.

5.1.35.1 Find the foot of the perpendicular from $P = (0, 2, 3)$ to $\frac{-x-3}{-5} = \frac{1-y}{-2} = \frac{3z+12}{9}$ and find its length.

Solution:
The line passes through $A = (-3, 1, -4)$ with direction $\mathbf{d} = (5, 2, 3)$. The projection parameter is $t = \frac{\mathbf{d}\cdot(P-A)}{|\mathbf{d}|^2} = 1$, so the foot is $H = A + \mathbf{d} = (2, 3, -1)$. Therefore $PH = \sqrt{(-2)^2 + (-1)^2 + 4^2} = \sqrt{21}$. The code plots the line and perpendicular.

(1.1.9)

codes/5.1.35.1.py

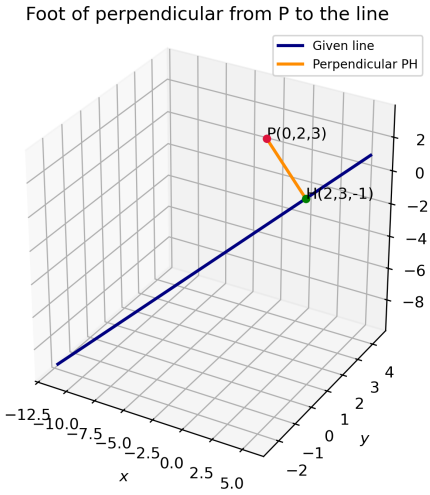


Fig. 12: Perpendicular from P to the given line.

5.1.35.2 The shortest distance between $\mathbf{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\mathbf{r} = (p\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$ is $3/\sqrt{2}$. Find p.

Solution:
The direction vectors have cross product $(-3, 0, 3)$. The skew-line distance is $\frac{|(Q-P)\cdot(\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|} = \frac{|p+1|}{\sqrt{2}}$. Equating this to $3/\sqrt{2}$ gives $|p+1| = 3$, so $\boxed{p = 2 \text{ or } -4}$. The code plots both configurations and evaluates the distance condition.

(1.1.9)

codes/5.1.35.2.py

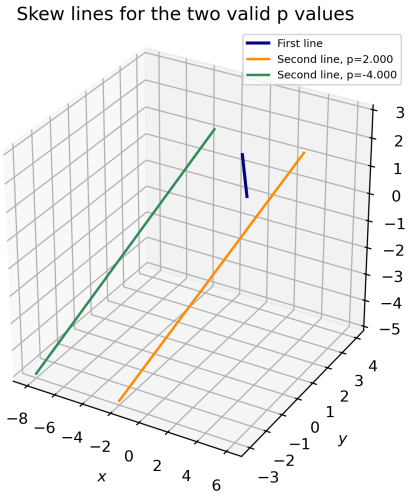


Fig. 13: The skew-line pair for each valid value of p.

5.1.36 Let $A = \{G_1, G_2, G_3, G_4\}$ and $B = \{B_1, B_2, B_3\}$. Find (i) the number of relations from A to B; (ii) whether $R = \{(G_1, B_1), (G_2, B_2), (G_3, B_2), (G_4, B_3), (G_1, B_2)\}$ is an injective function; (iii)(a) whether "students from the same colony" is an equivalence relation, or (iii)(b) whether a bijection $B \to A$ is possible.

Solution:
There are $2^{4\cdot 3} = 4096$ relations. In the listed R, G_1 is paired with two elements, so R is not a function and hence is not an injective function. "Being from the same colony" is reflexive, symmetric and transitive, so it is an equivalence relation. A bijection from B to A is impossible because $|B| = 3$ and $|A| = 4$. The C program prints these results.

(1.1.11)

codes/5.1.36.c

5.1.37 Vaccines A_1, A_2, A_3 are given to $25\%, 35\%, 40\%$ of the population. Their respective protection probabilities are $60\%, 55\%, 50\%$. Find (i) the probability an A_2 recipient gets infected; (ii) the probability a randomly chosen person is protected; (iii)(a) the probability an infected person had A_1, or (iii)(b) the probability an uninfected person had A_3.

Solution:
(i) $P(I \mid A_2) = 1 - 0.55 = 0.45$. (ii) $P(P) = 0.25(0.60) + 0.35(0.55) + 0.40(0.50) = 0.5425 = 217/400$, so $P(I) = 183/400$.

(1.1.6)

(iii)(a) $P(A_1 \mid I) = \frac{0.25(0.40)}{0.4575} = 40/183$. (iii)(b) $P(A_3 \mid P) = \frac{0.40(0.50)}{0.5425} = 80/217$.